

# THE SOLVE LAB

Everything a camera solve carries, made visible — the teaching workflow.

FILE `atlas_solve_lab_coastal_alley_workflow.json` VERIFIED 2026-07-16

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## THE STORY

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Before projecting a single pixel, you need to trust the solve. This workflow takes one plate with strong architectural lines — the classical vanishing-point solver's home turf — and lays out every diagnostic Atlas can produce: the VP fan-lines drawn over the photo, the horizon line with its confidence, ground and sky masks, an analytic ground depth map, and the decomposed camera numbers (focal, FOV, height, principal point).

It then walks the three-tier metric-scale ladder in one graph: a known-size reference object (a door, drawn as a bounding box) is tier 1; VLM-detected scale cues feed the same tier but only apply when you flip confirm ON; the assumed 1.6 m eye height is the flagged last resort. Finally, the solve is exported to JSON and RELOADED — the viewport renders the reloaded solve, proving the round-trip is lossless — alongside an artist-guided solve from explicit line groups and the legacy path-based node kept in a clearly-labeled corner.

## IN THE VIEWPORT — PROJECTED VS. THE GREY MESH

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#### SOURCE PLATE

coastal\_alley\_for\_vlm.png · 1408×768 — the single still photograph everything below is reconstructed from.



#### PROJECT OFF

The same geometry as a grey mesh, recovered view — what the depth-derived surfaces actually are.



#### GREY · ORBIT LEFT

Orbited off-axis: the reconstructed cone. Coverage ends where the camera never saw.



#### SOLVE DIAGRAM

 overlay: VP fan-lines, horizon and ground — the classical solve's evidence, drawn on the plate.

## NODES ON STAGE

AtlasSolveFromImage

AtlasVPVisualization

AtlasHorizonMask

AtlasGroundMask

AtlasGroundDepthMap

AtlasDecomposeSolve

AtlasDecomposeCamera

AtlasReferenceScaleSolve

AtlasVLMScaleCues

AtlasApplyScaleReferences

AtlasConstrainedSolve

AtlasExportSolveJSON

AtlasLoadSolveJSON

AtlasDepthAnything

AtlasLoadImageSolveCamera

## RETUNE ON YOUR PLATE

- ▶ **The reference bounding box is per-plate.** AtlasReferenceScaleSolve's bbox\_x0/y0/x1/y1 are pixel coordinates for THIS door. New plate = redraw the box and pick the right reference\_id (door / person / car).
- ▶ **The constraints JSON is per-plate too.** AtlasConstrainedSolve traces this alley's line groups; re-author the lines for a new photo or bypass the node.
- ▶ **VP solves need strong straight lines.** On organic or AI-generated plates the classical detector finds nothing — switch to AtlasLearnedSolveFromImage (GeoCalib).
- ▶ **VLM provider settings.** AtlasVLMScaleCues defaults to lmstudio :1234; point provider/base\_url at your own server, or 'openai' for a cloud endpoint.

## RUN-VERIFIED

**RUN** ✓ VP solve recovered an 85° hFOV; the door reference set camera height to 1.02 m; guided solve confidence 0.7. The VLM returned zero scale cues on this plate — and the confirm-gated apply correctly adopted nothing, which is the designed behavior.

## REQUIREMENTS

Atlas + [neural] · VLM optional (fails soft) · no gates

